

Polynomials

Problem 1. Let a, b, c, d be real numbers which are not all zero. Prove that the roots of the polynomial $x^6 + ax^3 + bx^2 + cx + d$ cannot all be real numbers.

Problem 2. Find all polynomials $P(x)$, $Q(x)$ and $R(x)$ such that

$$P(x) - Q(x) = R(x) \left(\sqrt{P(x)} + \sqrt{Q(x)} \right).$$

Problem 3. Find all polynomials $P(x)$ with integer coefficients such that $P(n) \mid 2^n + 3^n$ for all positive integers n .

Problem 4. Find all polynomials $P(x)$ with real coefficients satisfying the following condition: For any real number x , $P(x)$ is an integer if and only if x is an integer.

Problem 5. Find all polynomials $P(x)$ with integer coefficients satisfying the following condition: There exists an arithmetic progression $\{a + nd\}_{n \in \mathbb{Z}}$ such that $P(x) \neq a + nd$ for any integers x and n .

Problem 6. Let a and b be real numbers satisfying $b \geq a > 0$. Find all real numbers $x_1, x_2, \dots, x_n$ (in terms of a and b) such that $x_j^2 + 2ax_j + b^2 = x_{j+1}$ for $j = 1, 2, \dots, n$, where $x_{n+1} = x_1$.

Problem 7. Find all polynomials $P(x)$ with integer coefficients such that $P(n)$ is a palindrome number for every positive integer n . (A palindrome number is a positive integer that reads the same from left to right and from right to left.)

Problem 8. Find all pairs (m, n) of positive integers such that there exist two monic polynomials $P(x)$ and $Q(x)$ with $\deg P = m$ and $\deg Q = n$ satisfying $P(Q(t)) \neq Q(P(t))$ for all real numbers t .